Paul Platzer

PhD

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Post-doc (Lab-STICC)

2026-now	**Analysis of precursors of extreme events in the North-Atlantic ocean**
Institution	IMT-Atlantique, Plouzané
Lab-team	Lab-STICC, OSE
Supervisors	Pierre Tandeo, Florian Sévellec
Collaborators	Kinya Toride, Pierre Ailliot, Philippe Naveau
Objective	Analyze decadal ocean precursors of exceptional basin-wide and season-lasting sea-surface-temperature events in the North-Atlantic ocean (such as 2023-2024) using observational data, coupled-model ensemble climate projections, and analog methods.

Post-doc (LOPS-SIAM)

2021-2025	**Analog-based inference in atmosphere–ocean dynamics: metrics, uncertainty, and local geometry**
Institution	Ifremer, Plouzané
Lab-team	LOPS-SIAM
Supervisor	Bertrand Chapron (LOPS-SIAM)
Collaborators	Pierre Tandeo (IMT-A), Pierre Ailliot (UBO), Arthur Avenas (ESA), Alexis Mouche (LOPS), Léo Vinour (Hokkaido University), Gabriele Messori (Uppsala University)
First part	*Statistical relationship between storm surge and large-scale circulation*. 19th century-old extreme storm surges are used to complement sea-level pressure observations for the estimation of intense storms. (**1 journal paper**)
Second part	*Effect of unresolved scales on analogue forecasting ensembles*. The numerical study of an idealized system shows that analogue forecasting uncertainties depend on unresolved scales for long-term forecasts and analogue-target distances for short-term forecasts. (**1 conference paper**)

Third part *Understanding variations of local dimensions and applications to weather-regimes*. First, the transitions between North-Atlantic weather regimes are studied through the lens of indicators derived from dynamical systems theory and extreme value theory. Then, we derive analytical formulas for the density-based variations of local dimension estimates of random variables possessing an absolutely continuous probability density function. Finally, we develop a null-hypothesis test to asses density-effects on weather regime dimension estimates. (**1 conference paper**, **2 journal papers**)

Fourth part *Distance learning for analogue methods*. We develop an algorithm to optimize the distance used in analogue methods in order to minimize the continuous-ranked probability score of analogue ensemble methods. (**1 journal paper**)

Ph.D. (LSCE, Lab-STICC, FEM)

2017-2020 **Forecasts of dynamical systems from analogs: Applications to geophysical variables with a focus on ocean waves** – `https://tel.archives-ouvertes.fr/tel-03185865/`

Structures *Principal:* LSCE (Paris) – *Secondary:* Lab-STICC (Brest) & FEM (Brest)

Supervisors Pascal Yiou (LSCE), Philippe Naveau (LSCE), Pierre Tandeo (IMT-A), Thierry Chonavel (IMT-A), Jean-François Filipot (FEM)

First part *Forecast of extreme ocean surface gravity waves* – First, *in situ* data from wave boys in the Iroise Sea is studied (not shown in thesis), and analogue forecasts are performed (thesis chapter III.3.4). Then, a new physics-based methodology for the forecasts of extreme ocean surface gravity waves is proposed (paper below and thesis chapter IV).

Second part *Theoretical findings on analogue methodology* – First, analogue forecasting is linked to the system's local dynamics and tested on idealized dynamical systems (see paper below and thesis chapter III). Then, the distribution of analogue-to-target distances is evaluated using local dimensions and catalog size, borrowing from dynamical systems theory and extreme value theory (see paper below and thesis chapter II). Finally, analogue data assimilation is tested on idealized univariate heavy-tailed random variables (thesis chapter V).

Defense report "Paul Platzer a fait une présentation claire et très pédagogique d'une sélection judicieuse des nombreux résultats de sa thèse, portant sur divers aspects de la prédiction de systèmes dynamiques par la méthode des analogues. Le jury a apprécié la transdiciplinarité des travaux. Dans son exposé, Paul Platzer a abordé des problèmes théoriques et les a illustré par des applications sur des données réelles, afin de mettre en évidence leurs perspectives pratiques. Paul Platzer a donné des réponses détaillées et argumentées aux questions variées du jury. La discussion qui a suivi la présentation a montré sa grande maîtrise de divers domaines, sa capacité à synthétiser des questions complexes et sa maturité scientifique. Le jury est convaincu que Paul Platzer a de nombreuses qualités qui feront de lui un excellent chercheur et enseignant-chercheur. "

Project leading

DYNADISTNA (2027) *Dynamical distance optimization for basin-scale North Atlantic sea-surface temperature jump predictability*. Project funded by ISblue (6k€), including an internship, a seminar, and covering travel expenses. The objective is to improve our understanding of oceanic drivers of basin-scale marine heatwaves and to develop a new methodology for optimizing distances in analog methods.

Journal publications

Platzer, P., Avenas, A., Chapron, B., Drumetz, L., Mouche, A., Tandeo, P., Vinour, L. (2025). Distance Learning for Analog Methods. *Monthly Weather Review 153(10)*

Platzer, P., Chapron, B., Messori, G. (2025): Disentangling density and geometry in weather regime dimensions using stochastic twins. *npj Climate and atmospheric sciences*

Platzer, P., Chapron, B. (2025): Density-Induced Variations of Local Dimension Estimates for Absolutely Continuous Random Variables. *Journal of Statistical Physics*

Platzer, P., Ailliot, P., Chapron, B., & Tandeo, P. (2024). Could old tide gauges help estimate past atmospheric variability? *Climate of the Past*, 20, 2267–2286.

Avenas, A., Chapron, B., Mouche, A., **Platzer, P.**, & Vinour, L. (2024). Revealing short-term dynamics of tropical cyclone wind speeds from satellite synthetic aperture radar. *Scientific Reports*, 12808.

Platzer, P., Yiou, P., Naveau, P., Filipot, J. F., Thiébaut, M., & Tandeo, P. (2021). Probability distributions for analog-to-target distances. *Journal of the Atmospheric Sciences*, 78(10), 3317-3335.

Platzer, P., Yiou, P., Naveau, P., Tandeo, P., Filipot, J. F., Ailliot, P., & Zhen, Y. (2021). Using local dynamics to explain analog forecasting of chaotic systems. *Journal of the Atmospheric Sciences*, 78(7), 2117-2133.

Varing, A., Filipot, J. F., Delpey, M., Guitton, G., Collard, F., **Platzer, P.**, Roeber, V. & Morichon, D. (2021). Spatial distribution of wave energy over complex coastal bathymetries: Development of methodologies for comparing modeled wave fields with satellite observations. *Coastal Engineering*, 169, 103793.

Platzer, P., Filipot, J. F., Naveau, P., Tandeo, P., & Yiou, P. (2020). Wave group focusing in the ocean: estimations using crest velocities and a Gaussian linear model. *Natural Hazards*, 104(3), 2431-2449.

Peer-reviewed conference publications

Platzer, P., Chapron, B. (2024). The Effects of Unresolved Scales on Analogue Forecasting Ensembles. *Stochastic Transport in Upper Ocean Dynamics III*, 223.

Platzer, P., Chapron, B., & Tandeo, P. (2023). Dynamical Properties of Weather Regime Transitions. *Stochastic Transport in Upper Ocean Dynamics II*, 223.

Platzer, P., Yiou, P., Tandeo, P., Naveau, P., & Filipot, J. F. (2019). Predicting analog forecasting errors using dynamical systems. *9th Workshop on Climate Informatics*.

Selection of oral presentations

LOPS seminar, 2025 (*slides*)	https://paulplatzer.wordpress.com/wp-content/uploads/2025/04/presentation_lops-1.pdf
Climat et Impacts, 2024 (*slides*)	https://paulplatzer.wordpress.com/wp-content/uploads/2025/01/presentation_climatimpacts_platzer_update.pdf
Sea-Level Workshop, 2022 (*slides*)	https://paulplatzer.files.wordpress.com/2023/05/slw2022___preliminary_study_on_using_tide_gauge_measurements_to_reduce_uncertainty_in_historical_atmospheric_reanalysis.pdf
PhD, 2020 (*slides & video*)	https://paulplatzer.files.wordpress.com/2021/10/defense_platzer_2020.pdf | https://www.youtube.com/watch?v=SkHceNLNyrY

Education

2020	**Ph.D.**, *LSCE & IMT-A & FEM*, Paris and Brest, France *(see above)*.
2016	**M.S. Fluid Mechanics**, *(+ theoretical physics) l'X & UPMC*, Paris.
2015	**M.S. (1st year) Theoretical Physics**, *(+ geophysics) ENS Ulm*, Paris.
2014	**B.S. Theoretical Physics**, *(+ mathematics & geophysics) ENS Ulm*, Paris.
2010-2013	**Classe préparatoire, Mathematics and Physics**, *Lycée Michelet*, Vanves.

Internship experience

2016 (6m)	**3D Numerical Simulations of an Aerodynamic Condensation Trail**
Supervisor	Weeded Ghedhaïfi, *DEFA (ONERA)*, Palaiseau, France
Description	Numerical study of the formation and growth of aircraft condensation trails.
2015 (6m)	**Perturbed Cloud Microphysics due to Ice Nuclei**
Supervisors	Corinna Hoose and Marco Paukert, *IMK*, Karlsruhe, Germany
Description	Numerical study the impact of background desert dust aerosol concentration on the growth of a cold front through numerical simulations of the COSMO-model.
2014 (1m)	**Impact of a grating optical filter on infrared scintillometre measurements**
Supervisor	Jean-Martial Cohard, *LTHE (UJF)*, Grenoble, France
Description	Experimental evaluating of an optical filter for scintillometer heat flux measurements.

Teaching & supervising

2024–2025	**Last year student's internship**, *Statistical Analysis of temperature and salinity data distributions for the improvement of Quality Control (QC) procedures of COPERNICUS MARINE SERVICE*, Nour Abboud, POKaPOK, Brest, France. Co-supervisors: Jérôme Gourrion, Said Ouala
2023–2024	**Last year student's project**, *Révéler les tempêtes du XIXème siècle à partir de données marégraphiques par apprentissage*, Lorène Pourias, Hortense Ronzani, IMT-Atlantique, Brest, France. Co-supervisor: Pierre Tandeo

2022 **Pre-PhD research engineer**, *Atmospheric analogs for upwelling variability in St-Helena Bay*, Raquel Flügel, Ifremer, Brest, France.
Co-supervisors: Anne-Marie Tréguier, Steven Herbette, Bertrand Chapron

2021–2022 **Last year student's project**, *A Markov Chain Model for European Weather Prediction*, Loïc Turounet, Sami Rmili, Patrick Njeukam Njeumen, Joséphine Schmutz, Paul Muller, IMT-Atlantique, Brest, France.
Co-supervisor: Pierre Tandeo

2018–2019 **Last year student's project**, *Analog Forecasting combined with Markov-chains*, Adrien Mounier.
Co-supervisors: Pierre Tandeo, Alex Ayet

2018–2019 **Lecture courses and tutorials, engineering school (42 hours)**, First-year students, general and applied mathematics, IMT-Atlantique, Brest, France.

2013–2017 **Oral assessor in classes préparatoires (≈ 600 hours)**, bachelor level oral examinations ('colles'), general mathematics, physics, chemistry, Île de France.

Computer skills

Python (`scikit-learn`, `xarray`, `pandas`, `PyTorch`), R, Gimp, Inkscape, Unix.

Languages

French: native. English: fluent. German: B2.